Skylar: Troubleshooting Data Flow

Guide	Skylar
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Examples of end user experience that might indicate an issue with Skylar data flow include things like:

- Reporting data in Superset/ODBC is out of date
- Anomaly Charts in Anomaly Detection aren't showing predictive ranges

Examples like this might be service specific to the artifact which is presumably missing, but this guide is meant to give a typical workflow for where to start and how to proceed during triaging and debugging efforts.

Troubleshooting Workflow

A good first step when investigating a supposed data flow issue is to check ClickHouse, since it is ultimately the backbone of Skylar and provides a key inflection point on whether to focus efforts upstream or downstream from the main landing tables for data coming into Skylar. If data is not making it to the landing tables, the problem is probably upstream of ClickHouse. If data is making it to the landing tables, the problem is probably downstream.

Steps

Step 1: Check ClickHouse health
Step 2: Check critical tables

Step 3: Check upstream
Step 4: Check downstream

Step 1: Check ClickHouse health

Check ClickHouse: Skylar: Troubleshooting ClickHouse Availability

Step 2: Check critical tables

For Metrics (e.g., Perf Dyn App data via Data Pull)

Check the main landing table for SL1 exported metrics into ClickHouse: `<tenant>.otel_metrics_gauge`. If data is not making it here, we know to focus efforts upstream from ClickHouse (otc-router-collector, SL1 stack health and configuration, or possibly some networking configuration external to Skylar). Otherwise, we know to look downstream.

The following query will report the most recent timestamp of data that exists per datatype in the last 24 hours (replace `<tenant>` with target tenant database):

```
SELECT ResourceAttributes['sl.datatype.name'] as datatype, max(TimeUnix), NOW() FROM
<tenant>.otel_metrics_gauge WHERE TimeUnix >= NOW() - INTERVAL 24 HOUR GROUP BY datatype
ORDER BY datatype ASC;
```

Metrics are tagged with the `sl.datatype.name` ResourceAttribute. An empty value for this will align with the basic host information metrics that the `sl-otelcol` exporter on SL1 will always export, separate from Data Pull data.

<u>First, look at the empty string datatype:</u>

- **If there is a large gap between the two timestamps:** something is wrong with the export pathway as a whole and the issue is probably upstream from ClickHouse. You can likely stop focusing on ClickHouse as the culprit, and look upstream (go right to step 3!)
- **If the two timestamps are within a minute or so of each other**: then there is recent exported data. In this case, the main SL1 → Skylar connection has enough visibility for basic export to operate just fine and should rule out any network issues. It also means the `otc-router-collector` is probably ok as it's responsible for routing data from SL1 → ClickHouse.

Next, if the empty datatype is recent, then evaluate any datatypes of interest in the query results from above. If the complaint is stale dynamic app data in reporting for example, when was the last time the `dynamic_app` datatype had data exported to the `otel_metrics_gauge` landing table? If there is recent data which is apparently not

reflected downstream, then we can start to consider other services and ClickHouse tables downstream from the landing table. Go to step 4.

If metrics looks good, check metadata to make sure there is a recent active snapshot within the last few hours.

```
select *, NOW() from metadata_snapshot_inventory where status='active' order by timestamp desc limit 1;
```

No active snapshot means there's an SL1 Metadata Exporter problem since it won't complete. In which case, use this page: Skylar: Troubleshooting Metadata Export

Step 3: Check upstream

If step 2 suggested the problem is upstream, continue here; otherwise go to step 4.

Some things to check upstream:

- Check `otc-router-collector` pod health in Skylar.
- Check `sl-otelcol` health and configuration on SL1 stack.
- Check `Data Pull` health on SL1 stack.
- Check SL1 organizational filters (are the expected organizations enabled for Skylar?)

Step 4: Check downstream

If the data flow issue is reporting/superset/odbc data related:

- Go here: Skylar: Troubleshooting Missing Reporting Data

If the data flow issue is missing predictions/ranges:

- Go here: Skylar: Troubleshooting Anomaly Detection

If the data flow issue is missing Predictive Alerts / Charts:

- Go here: Skylar: Troubleshooting Predictive Alerts